

Nikhila Maddhikuntla

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Summary

Recent IT graduate skilled in AI vision systems, Python, and OpenCV. Experienced in deploying AI models, configuring hardware, and troubleshooting. Passionate about solving real-world challenges in industrial settings. Strong problem-solver, adaptable, and eager to contribute to cutting-edge AI projects in the automotive industry.

Work Experience

Kgate Technologies, Inc. | Software Engineer Internship Oct 2024 - Present

- Developed full-stack applications with React, JavaScript, Java and Python, optimizing performance and reducing load times by 20%.
- Designed and integrated RESTful APIs, enhancing scalability and enabling third-party service integrations
- Collaborated in Agile sprints with cross-functional teams to deliver high-quality software on time.
- Used AWS EC2/S3 for cloud deployment, Git for version control, and improved code quality through unit testing and refactoring.

Swecha | Full Stack Development Internship Jun 2022 - Nov 2022

- Contributed to the development of user-focused web applications, utilizing machine learning for enhanced analytics.
- Executed all SDLC phases, including requirements gathering, design, development, testing, and deployment.
- Developed and maintained web applications using Python (Django) for Back-End and JavaScript (React) for Front-End.
- Led the development of a scalable web application using React and Node.js, improving user engagement by 40%.

Projects

AI-Powered Motion Detection System for Industrial Environments | Personal Project Feb 2025 - Mar 2025

- Developed and deployed a real-time motion detection system using OpenCV and AI-based image processing to monitor industrial production lines.
- Configured industrial cameras, lighting, and edge computing units for optimal performance.
- Collected and annotated data to enhance algorithm accuracy, reducing false positives by 30%.
- Technologies: OpenCV, Python, MOP2, edge computing, industrial cameras.

Stock Investment Advisor Web Application | Personal Project Jan 2025 - Feb 2025

- Real-Time Stock Price Fetching: Utilized the Alpha Vantage API to retrieve intraday stock prices for user-provided stock symbols.
- AI-Powered Investment Advice: Integrated OpenAI's GPT-4 to analyze stock prices and provide actionable investment advice with explanations.
- User-Friendly Interface: Designed a responsive and intuitive web interface using HTML, CSS, and Bootstrap for seamless user interaction.
- Error Handling: Implemented robust error handling for API responses, ensuring users receive clear feedback for invalid inputs or API limitations.
- Secure API Key Management: Used environment variables to securely manage and access API keys, adhering to best practices for sensitive data handling.
- Technologies: Python, Flask (Backend); Alpha Vantage API (stock data), OpenAI GPT-4 (AI advice); HTML, CSS, Bootstrap (Frontend); Requests (API calls), python-dotenv (environment variables).

Education

Pittsburg State University Jan 2023 - May 2024

Master of Science, Information Technology

- **Achievements:** GPA: 3.5

Nalla Malla Reddy Engineering College Aug 2018 - May 2022

Bachelor of Technology, Electronics

- **Achievements:** GPA: 3.2

Skills

- **Programming :** Python, SQL, Java, API Integration, Data Preprocessing, Model Deployment
- **AI & NLP:** Prompt Engineering, Fine-Tuning Large Language Models (LLMs), Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), AI Agents
- **Web Development:** RESTful API, React.js, Next.js, HTML, Assembly
- **Tools & Frameworks:** OpenAI GPT, ChatGPT, Hugging Face Transformers, TensorFlow, PyTorch
- **Backend Technologies:** Node.js, Flask
- **Cloud & Deployment:** Azure, AWS (EC2, S3), Cloud Deployment
- **Software Development:** Object-Oriented Programming (OOP), Data Structures
- **Soft Skills:** Team collaboration, Communication, Adaptability, Problem Solving, Leadership, Independence

Certifications

- Learning SQL Programming - October 2024